

```

<!DOCTYPE html>

<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Under Pressure – Northern Shrimp & A Changing Ocean</title>
<link href="https://fonts.googleapis.com/css2?
family=Playfair+Display:ital,wght@0,400;0,700;1,400&family=DM+Sans:wght@300;400;500
rel="stylesheet">
<style>
  :root {
    --deep: #040d1a;
    --abyss: #071428;
    --mid: #0a2240;
    --surface: #0e3a6e;
    --foam: #a8d4f5;
    --biolume: #4ff0c0;
    --acid: #f7c948;
    --warm: #f06a45;
    --text: #d4eaf7;
    --muted: #7aaabf;
  }

  - { margin: 0; padding: 0; box-sizing: border-box; }

html { scroll-behavior: smooth; }

body {
background: var(-deep);
color: var(-text);
font-family: 'DM Sans', sans-serif;
overflow-x: hidden;
}

/* — ANIMATED OCEAN BACKGROUND — */
.ocean-bg {
position: fixed;
inset: 0;
z-index: 0;
overflow: hidden;
pointer-events: none;
}
.wave-layer {

```

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position: absolute;
width: 200%;
height: 100%;
background: radial-gradient(ellipse at 20% 50%, #0a2a5e22 0%, transparent 60%),
radial-gradient(ellipse at 80% 30%, #06204a33 0%, transparent 50%);
animation: drift 20s ease-in-out infinite alternate;
}
.wave-layer:nth-child(2) { animation-duration: 27s; animation-direction:
alternate-reverse; opacity: 0.6; }
.wave-layer:nth-child(3) { animation-duration: 35s; opacity: 0.4; }
@keyframes drift {
from { transform: translateX(0) translateY(0); }
to   { transform: translateX(-8%) translateY(3%); }
}
.particle {
position: absolute;
border-radius: 50%;
background: var(-biolume);
opacity: 0;
animation: float-up linear infinite;
}
@keyframes float-up {
0%   { opacity: 0; transform: translateY(0) scale(1); }
10%  { opacity: 0.4; }
90%  { opacity: 0.15; }
100% { opacity: 0; transform: translateY(-100vh) scale(0.5); }
}

/* — LAYOUT — */
section {
position: relative;
z-index: 1;
min-height: 100vh;
display: flex;
flex-direction: column;
justify-content: center;
padding: 6rem 2rem;
}
.container { max-width: 900px; margin: 0 auto; width: 100%; }

/* — HERO — */
#hero {
text-align: center;
padding-top: 10rem;
padding-bottom: 4rem;
}
.hero-eyebrow {

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font-size: 0.75rem;
letter-spacing: 0.3em;
text-transform: uppercase;
color: var(-biolume);
margin-bottom: 1.5rem;
opacity: 0;
animation: fadeUp 1s 0.3s forwards;
}

h1 {
font-family: 'Playfair Display', serif;
font-size: clamp(3rem, 8vw, 6rem);
line-height: 1.05;
margin-bottom: 1.5rem;
opacity: 0;
animation: fadeUp 1s 0.5s forwards;
}

h1 em { color: var(-biolume); font-style: italic; }

.hero-sub {
font-size: 1.1rem;
color: var(-muted);
max-width: 560px;
margin: 0 auto 3rem;
line-height: 1.7;
opacity: 0;
animation: fadeUp 1s 0.7s forwards;
}

.scroll-hint {
display: flex;
flex-direction: column;
align-items: center;
gap: 0.5rem;
color: var(-muted);
font-size: 0.8rem;
letter-spacing: 0.15em;
text-transform: uppercase;
opacity: 0;
animation: fadeUp 1s 1.2s forwards;
}

.scroll-hint svg { animation: bounce 2s ease-in-out infinite; }

@keyframes bounce {
0%, 100% { transform: translateY(0); }
50%      { transform: translateY(6px); }
}

@keyframes fadeUp {
from { opacity: 0; transform: translateY(24px); }
to   { opacity: 1; transform: translateY(0); }
}
```

```

/* — SECTION HEADERS — */
.section-label {
font-size: 0.7rem;
letter-spacing: 0.3em;
text-transform: uppercase;
color: var(-biolume);
margin-bottom: 0.75rem;
}
h2 {
font-family: 'Playfair Display', serif;
font-size: clamp(1.8rem, 4vw, 2.8rem);
line-height: 1.2;
margin-bottom: 1.25rem;
}
h2 em { color: var(-foam); font-style: italic; }
.lead {
font-size: 1rem;
color: var(-muted);
line-height: 1.75;
max-width: 680px;
margin-bottom: 2rem;
}

/* — INTRO CARDS — */
.cards {
display: grid;
grid-template-columns: repeat(auto-fit, minmax(220px, 1fr));
gap: 1.25rem;
margin-top: 2.5rem;
}
.card {
background: rgba(14,58,110,0.25);
border: 1px solid rgba(168,212,245,0.1);
border-radius: 12px;
padding: 1.5rem;
backdrop-filter: blur(8px);
transition: border-color 0.3s, transform 0.3s;
}
.card:hover { border-color: rgba(79,240,192,0.35); transform: translateY(-3px); }
.card-icon { font-size: 1.8rem; margin-bottom: 0.75rem; }
.card h3 { font-size: 0.95rem; font-weight: 500; margin-bottom: 0.4rem; color:
var(-foam); }
.card p { font-size: 0.85rem; color: var(-muted); line-height: 1.6; }

/* — DIVIDER — */
.divider {

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width: 60px; height: 2px;
background: linear-gradient(90deg, var(-biolume), transparent);
margin: 2rem 0;
}

/* — INTERACTIVE SECTION — */
#interactive { padding-top: 4rem; padding-bottom: 4rem; }

.slider-panel {
background: rgba(7,20,40,0.7);
border: 1px solid rgba(168,212,245,0.12);
border-radius: 16px;
padding: 2rem 2.5rem;
backdrop-filter: blur(16px);
margin-bottom: 2rem;
}

.sliders-row {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 2.5rem;
margin-bottom: 1.5rem;
}

@media(max-width:600px){ .sliders-row { grid-template-columns: 1fr; } }

.slider-group label {
display: flex;
justify-content: space-between;
align-items: baseline;
margin-bottom: 0.6rem;
font-size: 0.85rem;
color: var(-muted);
text-transform: uppercase;
letter-spacing: 0.1em;
}

.slider-group label span {
font-family: 'Playfair Display', serif;
font-size: 1.3rem;
color: var(-text);
letter-spacing: 0;
text-transform: none;
}

#tempVal { color: var(-warm); }
#phVal   { color: var(-acid); }

input[type=range] {
-webkit-appearance: none;

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width: 100%;
height: 4px;
border-radius: 2px;
outline: none;
cursor: pointer;
}
#tempSlider {
background: linear-gradient(90deg, #0e3a6e, var(-warm));
}
#phSlider {
background: linear-gradient(90deg, var(-acid), #0e3a6e);
}
input[type=range]::-webkit-slider-thumb {
-webkit-appearance: none;
width: 18px; height: 18px;
border-radius: 50%;
background: white;
box-shadow: 0 0 10px rgba(255,255,255,0.4);
cursor: pointer;
}

.condition-badge {
text-align: center;
padding: 0.75rem;
border-radius: 8px;
font-size: 0.9rem;
font-weight: 500;
letter-spacing: 0.05em;
transition: all 0.4s;
margin-top: 0.5rem;
}

/* — VISUALIZATION GRID — */
.viz-grid {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 1.25rem;
}
@media(max-width:700px){ .viz-grid { grid-template-columns: 1fr; } }

.origin-card {
background: rgba(7,20,40,0.6);
border: 1px solid rgba(168,212,245,0.1);
border-radius: 14px;
padding: 1.5rem;
backdrop-filter: blur(8px);
transition: border-color 0.4s;

```

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}
.origin-card.active-origin { border-color: rgba(79,240,192,0.4); }

.origin-header {
display: flex;
justify-content: space-between;
align-items: center;
margin-bottom: 1rem;
}
.origin-name {
font-size: 0.8rem;
text-transform: uppercase;
letter-spacing: 0.15em;
color: var(-foam);
font-weight: 500;
}
.dap-count {
font-family: 'Playfair Display', serif;
font-size: 2rem;
font-weight: 700;
transition: color 0.4s;
line-height: 1;
}
.dap-label { font-size: 0.7rem; color: var(-muted); text-transform: uppercase;
letter-spacing: 0.1em; }

/* Protein network dots */
.network-viz {
position: relative;
width: 100%;
height: 120px;
margin: 1rem 0;
}
.network-viz svg { width: 100%; height: 100%; }

/* Bar showing relative disruption */
.disruption-bar-wrap {
margin-top: 0.5rem;
}
.disruption-bar-label {
display: flex;
justify-content: space-between;
font-size: 0.72rem;
color: var(-muted);
margin-bottom: 0.3rem;
}
.disruption-bar-track {

```

```

width: 100%;
height: 6px;
background: rgba(255,255,255,0.07);
border-radius: 3px;
overflow: hidden;
}
.disruption-bar-fill {
height: 100%;
border-radius: 3px;
transition: width 0.6s cubic-bezier(0.4,0,0.2,1), background 0.4s;
}

/* — INTERPRETATION BOX — */
.interpretation {
background: rgba(79,240,192,0.06);
border: 1px solid rgba(79,240,192,0.2);
border-radius: 12px;
padding: 1.25rem 1.5rem;
margin-top: 1.5rem;
font-size: 0.92rem;
line-height: 1.7;
color: var(-text);
transition: all 0.4s;
min-height: 80px;
}
.interpretation strong { color: var(-biolume); }

/* — EXPLAINER SECTION — */
#explainer { padding-top: 2rem; }
.explainer-grid {
display: grid;
grid-template-columns: 1fr 1fr;
gap: 1.5rem;
margin-top: 2rem;
}
@media(max-width:640px){ .explainer-grid { grid-template-columns: 1fr; } }

.explainer-block {
background: rgba(14,58,110,0.2);
border: 1px solid rgba(168,212,245,0.08);
border-radius: 12px;
padding: 1.5rem;
}
.explainer-block h4 {
font-family: 'Playfair Display', serif;
font-size: 1.1rem;
color: var(-foam);

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margin-bottom: 0.6rem;
}
.explainer-block p {
font-size: 0.87rem;
color: var(-muted);
line-height: 1.7;
}
.explainer-block .tag {
display: inline-block;
background: rgba(79,240,192,0.12);
color: var(-biolume);
font-size: 0.7rem;
padding: 0.2rem 0.6rem;
border-radius: 20px;
letter-spacing: 0.1em;
text-transform: uppercase;
margin-bottom: 0.7rem;
}

/* — WHAT'S AT STAKE — */
#stakes { padding-top: 2rem; }
.stakes-stat {
display: flex;
align-items: baseline;
gap: 0.75rem;
margin-bottom: 1.5rem;
}
.stakes-number {
font-family: 'Playfair Display', serif;
font-size: 3.5rem;
color: var(-biolume);
line-height: 1;
flex-shrink: 0;
}
.stakes-desc { font-size: 0.95rem; color: var(-muted); line-height: 1.6; max-width: 400px; }
.stakes-desc strong { color: var(-text); }

/* — FOOTER — */
footer {
position: relative;
z-index: 1;
text-align: center;
padding: 3rem 2rem;
border-top: 1px solid rgba(168,212,245,0.08);
color: var(-muted);
font-size: 0.8rem;

```

```

line-height: 1.8;
}
footer a { color: var(-biolume); text-decoration: none; }

/* — SCROLL ANIMATIONS — */
.reveal {
opacity: 0;
transform: translateY(30px);
transition: opacity 0.8s, transform 0.8s;
}
.reveal.visible {
opacity: 1;
transform: translateY(0);
}
</style>

</head>
<body>

<!-- Ocean background -->

<div class="ocean-bg">
  <div class="wave-layer"></div>
  <div class="wave-layer"></div>
  <div class="wave-layer"></div>
</div>

<!-- — HERO — -->

<section id="hero">
  <div class="container">
    <div class="hero-eyebrow">Science Communication · Marine Biology</div>
    <h1>Under<br><em>Pressure</em></h1>
    <p class="hero-sub">
      The Northern shrimp – backbone of one of the world's largest shellfish
      fisheries – is facing a changing ocean. But not all shrimp are the same. Explore
      how ocean acidification rewrites their molecular machinery, one population at a
      time.
    </p>
    <div class="scroll-hint">
      <span>Scroll to explore</span>
      <svg width="16" height="20" viewBox="0 0 16 20" fill="none">
        <path d="M8 0v16M1 10l7 8 7-8" stroke="#7aaabf" stroke-width="1.5" stroke-
        linecap="round" stroke-linejoin="round"/>
      </svg>
    </div>
  </div>
</div>

```

</section>

<!-- — INTRO — -->

<section id="intro">

<div class="container reveal">

<div class="section-label">The Species</div>

<h2>Pandalus borealis – More Than a Meal</h2>

<p class="lead">

The Northern shrimp is an ecological keystone and economic pillar of Atlantic Canada. It's also a living molecular recorder – and scientists are now reading what a changing ocean is writing inside its cells.

</p>

<div class="divider"></div>

<div class="cards">

<div class="card">

<div class="card-icon"></div>

<h3>Ecological Role</h3>

<p>A critical prey species for cod, redfish, whales, and seabirds across the North Atlantic.</p>

</div>

<div class="card">

<div class="card-icon"></div>

<h3>Economic Value</h3>

<p>The 3rd most lucrative fishery in Canadian waters – and one of the world's largest shellfish fisheries.</p>

</div>

<div class="card">

<div class="card-icon"></div>

<h3>Climate Threat</h3>

<p>Temperatures projected to rise ~4°C and ocean pH to drop 0.3 units by 2100 – in its core habitat.</p>

</div>

<div class="card">

<div class="card-icon"></div>

<h3>The Surprise</h3>

<p>Acidification – not warming – is the dominant driver of molecular disruption. And where a shrimp is from matters enormously.</p>

</div>

</div>

</div>

</section>

<!-- — INTERACTIVE — -->

<section id="interactive">

<div class="container">

```

<div class="reveal">
  <div class="section-label">Interactive Explorer</div>
  <h2>Adjust the Ocean.<br>Watch the <em>Proteome Respond.</em></h2>
  <p class="lead">
    Use the sliders below to simulate future ocean conditions. See how the
    number of disrupted proteins – and network complexity – changes across four
    Canadian shrimp populations.
  </p>
</div>

...

<div class="slider-panel reveal">
  <div class="sliders-row">
    <div class="slider-group">
      <label>Temperature <span id="tempVal">2°C</span></label>
      <input type="range" id="tempSlider" min="0" max="2" step="1" value="0">
      <div style="display:flex;justify-content:space-between;font-
size:0.72rem;color:var(--muted);margin-top:0.4rem;">
        <span>2°C (current)</span><span>6°C (+4°C)</span><span>10°C (+8°C)</span>
      </div>
    </div>
    <div class="slider-group">
      <label>Seawater pH <span id="phVal">7.75</span></label>
      <input type="range" id="phSlider" min="0" max="1" step="1" value="0">
      <div style="display:flex;justify-content:space-between;font-
size:0.72rem;color:var(--muted);margin-top:0.4rem;">
        <span>pH 7.75 (current)</span><span>pH 7.40 (2100)</span>
      </div>
    </div>
  </div>
  <div class="condition-badge" id="conditionBadge">
     Favourable conditions – close to current ocean state
  </div>
</div>

<div class="viz-grid reveal" id="vizGrid">
  <!-- Cards populated by JS -->
</div>

<div class="interpretation reveal" id="interpretation">
  Move the sliders to begin exploring how ocean conditions affect each
  population's proteome.
</div>
...

</div>
</section>

```

<!-- — EXPLAINER — -->

<section id="explainer">

<div class="container reveal">

<div class="section-label">The Science</div>

<h2>Why Does It Matter Which Population?</h2>

<p class="lead">

These shrimp look identical – but their molecular responses to stress are dramatically different, shaped by decades of acclimatisation to local ocean conditions.

</p>

<div class="explainer-grid">

<div class="explainer-block">

<div class="tag">St. Lawrence Estuary</div>

<h4>The Most Responsive</h4>

<p>SLE shrimp show 109 differentially abundant proteins under acidification – 18–36× more than any other population. Their molecular network is vast and highly interconnected, spanning metabolism, stress response, and ion homeostasis.</p>

</div>

<div class="explainer-block">

<div class="tag">Esquiman Channel</div>

<h4>Quietly Vulnerable</h4>

<p>EC shrimp show almost no molecular response – just 3 affected proteins. This silence may not mean resilience. With no suitable habitat to migrate into, these shrimp may be the most ecologically at risk.</p>

</div>

<div class="explainer-block">

<div class="tag">Eastern Scotian Shelf</div>

<h4>Immune Signal</h4>

<p>ESS shrimp showed just 6 disrupted proteins, but one key finding: enrichment in pathways for negative regulation of innate immune response – suggesting OA may suppress disease defense.</p>

</div>

<div class="explainer-block">

<div class="tag">NE Newfoundland Coast</div>

<h4>Minimal Remodelling</h4>

<p>NNC shrimp also show only 6 disrupted proteins with minimal network complexity – suggesting either low sensitivity to acidification or a reduced capacity to mount a plastic response.</p>

</div>

</div>

</div>

</section>

<!-- — STAKES — -->

```

<section id="stakes">
  <div class="container reveal">
    <div class="section-label">What's at Stake</div>
    <h2>The Bigger Picture</h2>
    <div class="divider"></div>
    <div class="stakes-stat">
      <div class="stakes-number">-0.3</div>
      <div class="stakes-desc"><strong>pH units</strong> projected to drop in key
habitats by 2100 – the same drop that triggers massive molecular disruption in SLE
shrimp.</div>
    </div>
    <div class="stakes-stat">
      <div class="stakes-number">109</div>
      <div class="stakes-desc"><strong>proteins disrupted</strong> in SLE shrimp
under acidification – including those governing energy metabolism, stress
response, and ion balance.</div>
    </div>
    <div class="stakes-stat">
      <div class="stakes-number">4×</div>
      <div class="stakes-desc"><strong>populations studied</strong> – each
responding differently. Uniform fisheries management may fail to protect the most
vulnerable.</div>
    </div>
    <p class="lead" style="margin-top:2rem;">
      Understanding <em>which</em> populations are most sensitive – and why – is
now essential for designing conservation strategies that can keep pace with a
rapidly acidifying ocean.
    </p>
  </div>
</section>

<footer>
  <p>Based on: Page et al. (2026) <em>Marine Pollution Bulletin</em> 225,
119152</p>
  <p style="margin-top:0.4rem;"><em>Pandalus borealis</em> proteome data from four
Atlantic Canadian populations exposed to ocean acidification and warming.</p>
</footer>

<script>
// — DATA from paper —
const origins = [
  {
    id: 'SLE',
    name: 'St. Lawrence Estuary',
    abbr: 'SLE',
    // DAPs for [2C, 6A, 10A] × [pH7.75, pH7.40]
  }
]

```

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// rows: temp index (0=2°C, 1=6°C, 2=10°C), cols: pH (0=7.75, 1=7.40)
daps: [[0, 109], [0, 72], [0, 85]],
// "network complexity" score 0-100 (relative, derived from paper's
description)
complexity: [[2, 95], [2, 80], [2, 88]],
color: '#4ff0c0',
note: {
  low: 'Under current conditions, SLE shrimp proteome is relatively stable.',
  high: 'Under acidification, SLE shrimp show <strong>widespread proteomic
disruption</strong> – 109 differentially abundant proteins spanning metabolism,
stress response, and ion homeostasis.'
}
},
{
  id: 'EC',
  name: 'Esquiman Channel',
  abbr: 'EC',
  daps: [[0, 3], [0, 2], [0, 3]],
  complexity: [[1, 8], [1, 6], [1, 7]],
  color: '#f7c948',
  note: {
    low: 'EC shrimp proteome remains stable under current conditions.',
    high: 'Minimal proteomic response (3 DAPs) – but this may signal
<strong>vulnerability rather than resilience</strong>, as EC shrimp have nowhere
to migrate.'
  }
},
{
  id: 'ESS',
  name: 'Eastern Scotian Shelf',
  abbr: 'ESS',
  daps: [[0, 6], [0, 4], [0, 5]],
  complexity: [[1, 18], [1, 12], [1, 15]],
  color: '#a8d4f5',
  note: {
    low: 'ESS shrimp show no significant proteomic response under current pH.',
    high: 'ESS shrimp show 6 disrupted proteins, with an enriched pathway for
<strong>negative regulation of innate immune response</strong> – a potential
vulnerability to disease.'
  }
},
{
  id: 'NNC',
  name: 'NE Newfoundland Coast',
  abbr: 'NNC',
  daps: [[0, 6], [0, 4], [0, 5]],
  complexity: [[1, 15], [1, 10], [1, 12]],

```

```

    color: '#f06a45',
    note: {
      low: 'NNC shrimp proteome is largely unaffected under current conditions.',
      high: 'NNC shrimp show 6 disrupted proteins – minimal molecular remodelling,
suggesting <strong>reduced sensitivity or limited plastic response</strong> to
acidification.'
    }
  }
];

```

```

const tempLabels = ['2°C', '6°C', '10°C'];
const pHLabels   = ['7.75', '7.40'];

```

```

function getState() {
  const t = parseInt(document.getElementById('tempSlider').value);
  const p = parseInt(document.getElementById('phSlider').value);
  return { t, p };
}

```

```

// Generate SVG network

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```

function buildNetwork(nodes, connections, color, animDelay) {
  const W = 200, H = 100;
  const pts = [];
  // Deterministically place nodes
  for (let i = 0; i < nodes; i++) {
    const seed1 = (i * 73 + 17) % 100;
    const seed2 = (i * 41 + 53) % 80;
    pts.push({ x: 10 + seed1 * 1.8, y: 10 + seed2 });
  }
  let svg = `<svg viewBox="0 0 ${W} ${H}" xmlns="http://www.w3.org/2000/svg">`;
  // Draw edges
  const edgeCount = Math.min(connections, Math.floor(nodes * 1.5));
  for (let e = 0; e < edgeCount && pts.length > 1; e++) {
    const i = e % pts.length;
    const j = (e * 3 + 2) % pts.length;
    if (i !== j) {
      svg += `<line x1="${pts[i].x}" y1="${pts[i].y}" x2="${pts[j].x}"
y2="${pts[j].y}"
      stroke="${color}" stroke-width="0.6" opacity="0.25"/>`;
    }
  }
  // Draw nodes
  pts.forEach((p, i) => {
    const r = i < 3 ? 3.5 : 2;
    const op = i < 5 ? 0.9 : 0.5;
    svg += `<circle cx="${p.x}" cy="${p.y}" r="${r}" fill="${color}"
opacity="${op}">

```



```

    <animate attributeName="opacity" values="${op};${op*0.4};${op}" dur="${2 +
(i%3)}s" repeatCount="indefinite" begin="${(i*0.15).toFixed(2)}s"/>
  </circle>`;
});
svg += '</svg>';
return svg;
}

```

```

function update() {
  const { t, p } = getState();

  // Update labels
  document.getElementById('tempVal').textContent = tempLabels[t];
  document.getElementById('phVal').textContent = phLabels[p];

  // Condition badge
  const badge = document.getElementById('conditionBadge');
  let badgeText, badgeBg;
  if (p === 0) {
    badgeText = '✅ Favourable conditions – close to current ocean pH (7.75)';
    badgeBg = 'rgba(79,240,192,0.1)';
    badge.style.color = 'var(--biolume)';
    badge.style.borderColor = 'rgba(79,240,192,0.2)';
  } else if (t === 0) {
    badgeText = '⚠️ Ocean acidification only – pH drops to 7.40 (projected 2100)';
    badgeBg = 'rgba(247,201,72,0.1)';
    badge.style.color = 'var(--acid)';
    badge.style.borderColor = 'rgba(247,201,72,0.25)';
  } else {
    badgeText = '🔴 Combined stressor – elevated temperature AND acidification';
    badgeBg = 'rgba(240,106,69,0.1)';
    badge.style.color = 'var(--warm)';
    badge.style.borderColor = 'rgba(240,106,69,0.25)';
  }
  badge.textContent = badgeText;
  badge.style.background = badgeBg;

  // Update origin cards
  const grid = document.getElementById('vizGrid');
  grid.innerHTML = '';

  origins.forEach(o => {
    const daps = o.daps[t][p];
    const comp = o.complexity[t][p];
    const maxDaps = 109;
    const pct = Math.round((daps / maxDaps) * 100);

```

```

// nodes / edges for network viz
const nodeCount = Math.max(3, Math.round(comp * 0.5));
const edgeCount = Math.max(0, Math.round(comp * 0.3));

const barColor = p === 0
  ? 'rgba(168,212,245,0.4)'
  : daps > 20 ? o.color : `${o.color}99`;

const card = document.createElement('div');
card.className = `origin-card ${daps > 10 ? 'active-origin' : ''}`;
card.innerHTML = `
  <div class="origin-header">
    <div>
      <div class="origin-name">${o.abbr}</div>
      <div style="font-size:0.72rem;color:var(--muted)">${o.name}</div>
    </div>
    <div style="text-align:right">
      <div class="dap-count" style="color:${daps > 10 ? o.color : 'var(--muted)'}">${daps}</div>
      <div class="dap-label">disrupted<br>proteins</div>
    </div>
    <div class="network-viz">${buildNetwork(nodeCount, edgeCount, o.color, 0)}
  </div>
  <div class="disruption-bar-wrap">
    <div class="disruption-bar-label">
      <span>Proteomic disruption</span>
      <span>${pct}% of max observed</span>
    </div>
    <div class="disruption-bar-track">
      <div class="disruption-bar-fill"
style="width:${pct}%;background:${barColor}"></div>
    </div>
  </div>
`;
grid.appendChild(card);
});

// Interpretation text
const interp = document.getElementById('interpretation');
if (p === 0) {
  interp.innerHTML = `<strong>Current ocean conditions (pH 7.75):</strong> All
four populations show minimal proteomic disruption – their molecular machinery is
operating close to baseline. This represents conditions shrimp have adapted to
over generations. Notice how few proteins are affected regardless of temperature,
confirming that <strong>warming alone does not drive significant proteomic
change</strong> in this species.`;

```

```

    } else if (t === 0) {
        interp.innerHTML = `Ocean acidification at 2°C: The contrast
between populations becomes stark. SLE shrimp show 109 disrupted
proteins – a massive molecular upheaval spanning energy metabolism, stress
response, and ion balance. Meanwhile, EC, ESS, and NNC shrimp show fewer than 6
disrupted proteins each. Same species. Completely different molecular stories.`;
    } else {
        interp.innerHTML = `Combined stressors (elevated temperature +
acidification): Even with warming added, acidification remains
the dominant driver of proteomic change. SLE shrimp continue to show
dramatically elevated disruption across multiple biological pathways. This
challenges the expectation that warming would be the primary stressor – at the
molecular level, pH tells a bigger story.`;
    }
}

// — INIT CARDS —
function initCards() {
    const grid = document.getElementById('vizGrid');
    grid.innerHTML = '';
    origins.forEach(o => {
        const card = document.createElement('div');
        card.className = 'origin-card';
        card.innerHTML = `
            <div class="origin-header">
                <div>
                    <div class="origin-name">${o.abbr}</div>
                    <div style="font-size:0.72rem;color:var(--muted)">${o.name}</div>
                </div>
                <div style="text-align:right">
                    <div class="dap-count" style="color:var(--muted)">0</div>
                    <div class="dap-label">disrupted<br>proteins</div>
                </div>
            </div>
            <div class="network-viz">${buildNetwork(3, 0, o.color, 0)}</div>
            <div class="disruption-bar-wrap">
                <div class="disruption-bar-label"><span>Proteomic disruption</span>
                <span>0%</span></div>
                <div class="disruption-bar-track">
                    <div class="disruption-bar-fill" style="width:0%;background:${o.color}">
            </div>
            </div>
            </div>
            </div>
        `;
        grid.appendChild(card);
    });
}

```

```

document.getElementById('tempSlider').addEventListener('input', update);
document.getElementById('phSlider').addEventListener('input', update);

// — PARTICLES —
function spawnParticles() {
  const bg = document.querySelector('.ocean-bg');
  for (let i = 0; i < 18; i++) {
    const p = document.createElement('div');
    p.className = 'particle';
    const size = 2 + Math.random() * 3;
    p.style.cssText = `
      width:${size}px;height:${size}px;
      left:${Math.random()*100}%;
      bottom:${Math.random()*30}%;
      animation-duration:${10+Math.random()*20}s;
      animation-delay:${Math.random()*15}s;
    `;
    bg.appendChild(p);
  }
}

// — SCROLL REVEAL —
const observer = new IntersectionObserver(entries => {
  entries.forEach(e => { if (e.isIntersecting) e.target.classList.add('visible');
});
}, { threshold: 0.12 });
document.querySelectorAll('.reveal').forEach(el => observer.observe(el));

spawnParticles();
initCards();
update();
</script>

</body>
</html>

```